

Applied Data Analysis
Winter 2025
Probability & Statistics
Due January 13, 2025, 5:30pm

For this assignment, clearly indicate your answers to each problem and submit a pdf to the GitHub Classroom repo here: <https://classroom.github.com/a/JWjVHc1r>

Pass: Complete 4 of the problems below completely correctly

1. Suppose we roll two dice, one blue and one red. Let B be the number shown on the blue die, and let R be the number shown on the red die. Further, let $X = B + R$ and $Y = |B - R|$. For example, if the blue die shows 5 and the red die shows 2, then $X = 7$ and $Y = 3$.
 - (a) Find $p_X(x)$ and $p_Y(y)$.
 - (b) Find $E[X]$ and $E[Y]$.
 - (c) Find $\text{Var}(X)$ and $\text{Var}(Y)$.
 - (d) Find $\text{Cov}(X, Y)$ and $\rho_{X,Y}$.
2. Suppose a jar has 1,000 quarters. There are 999 fair quarters, with “Heads” on one side, and there is 1 two-headed quarter, with “Heads” on both sides. You pull out a quarter at random, set it on the table, and the side facing up shows “Heads”. What is the probability that this is the two-headed quarter?
3. Of 30 students in a class, 10 are from Illinois. If 4 students are selected at random from the class, what is the probability that at least 3 are from Illinois?
4. The UChicago shuttle bus runs every 10 minutes, on average. When you go to the bus stop, what is the probability you will wait more than 5 minutes for the bus?
5. A machine stamps out widgets with thickness $X \sim \mathcal{N}(31, 0.25)$ which are used to build a machine.
 - (a) The machine being built requires widgets whose thickness is between 30 and 32 millimeters thick. What is the probability that a randomly selected widget can be used in the machine?
 - (b) If 100 widgets are chosen at random, what is the probability at least 95 can be used?